

JOINT LOW-RANK REPRESENTATION AND MATRIX COMPLETION UNDER A SINGULAR VALUE THRESHOLDING FRAMEWORK

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Overview

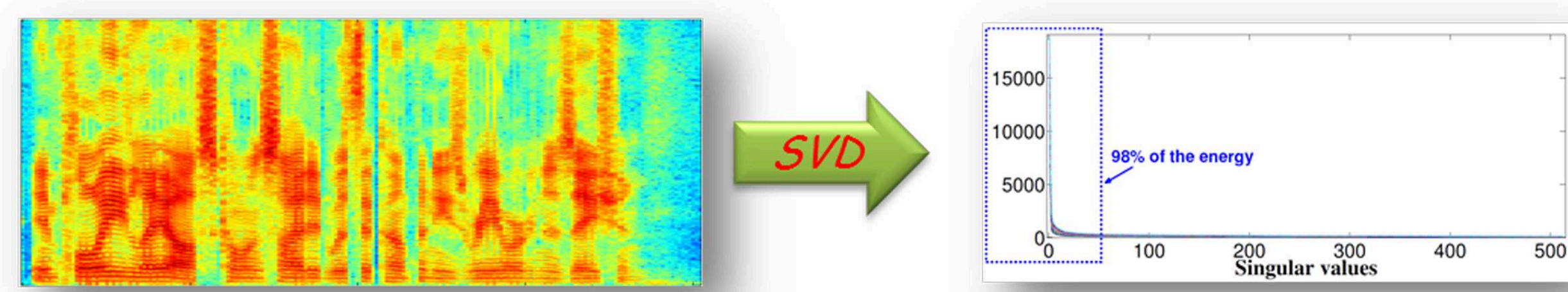
- Many real-world problems require the estimation of a matrix with missing entries (e.g., video denoising, audio bandwidth expansion, speech feature reconstruction, localization in WLANs, etc.)
- Adopt *matrix completion* to compute the missing entries in a partially observed data matrix
 - solve a rank minimization problem, where the *nuclear norm* is used as the convex surrogate of the rank function
- Nuclear norm minimization has been developed in parallel with matrix completion especially for subspace segmentation
 - seek for the *lowest rank representation* of a data matrix with respect to a collection of data drawn from a union of multiple subspaces

Goal: estimate the lowest rank representation of a data matrix with respect to a given basis or dictionary connected with a partially observed version of it under a singular value thresholding scheme

Low-rank matrix recovery



- YES!** ...this is what matrix completion does
- Core concept:** exploit low-rank behaviour of the original data matrix
- Ordered singular values** of the original data matrix



– *low-rankness assumption* $\implies$ the majority of singular values tend to zero

- Assumptions:**
 - observed entries of original data matrix $M \in \mathbb{R}^{n_1 \times n_2} \implies \mathcal{O}(nr \ln^2 n)$
 - $r = \text{rank}(M)$, $n = \max\{n_1, n_2\}$
 - known entries $M_{ij} \implies (i, j) \in \mathcal{I} \subset \{1, \dots, n_1\} \times \{1, \dots, n_2\}$
 - linear map $\mathcal{A} : \mathbb{R}^{n_1 \times n_2} \rightarrow \mathbb{R}^k \implies$ sampling operator recording a specific number of matrix entries ($k = |\mathcal{I}|$)
 - $\mathbf{b} := \mathcal{A}_T(M)$ contains the sample values extracted from M
 - $\mathcal{A}_T(X) \equiv \mathbf{A}\mathbf{x}$, $\mathbf{x} := \text{vec}(X)$, $\forall X \in \mathbb{R}^{n_1 \times n_2}$
 - $\mathbf{A} \in \mathbb{R}^{n_1 \times n_2}$, $\text{vec}(\cdot) : \mathbb{R}^{n_1 \times n_2} \rightarrow \mathbb{R}^{n_1 n_2 \times 1}$
 - adjoint operator $\mathcal{A}_T^*(\cdot) : \mathbb{R}^{k \times 1} \rightarrow \mathbb{R}^{n_1 \times n_2}$ with $k = |\mathcal{I}| \leq n_1 n_2$
 - maps a vector into a sparse matrix
 - $\mathcal{A}_T(\mathcal{A}_T^*(\mathbf{h})) = \mathbf{h} \forall \mathbf{h} \in \mathbb{R}^{k \times 1}$
- Recovery of the original matrix from the partially observed matrix by solving the *nuclear norm minimization problem* using the *Singular Value Thresholding (SVT)* algorithm

$$\min_X \tau \|X\|_* + \frac{1}{2} \|X\|_F^2 \quad \text{s.t. } \mathcal{A}_T(X) = \mathbf{b}$$

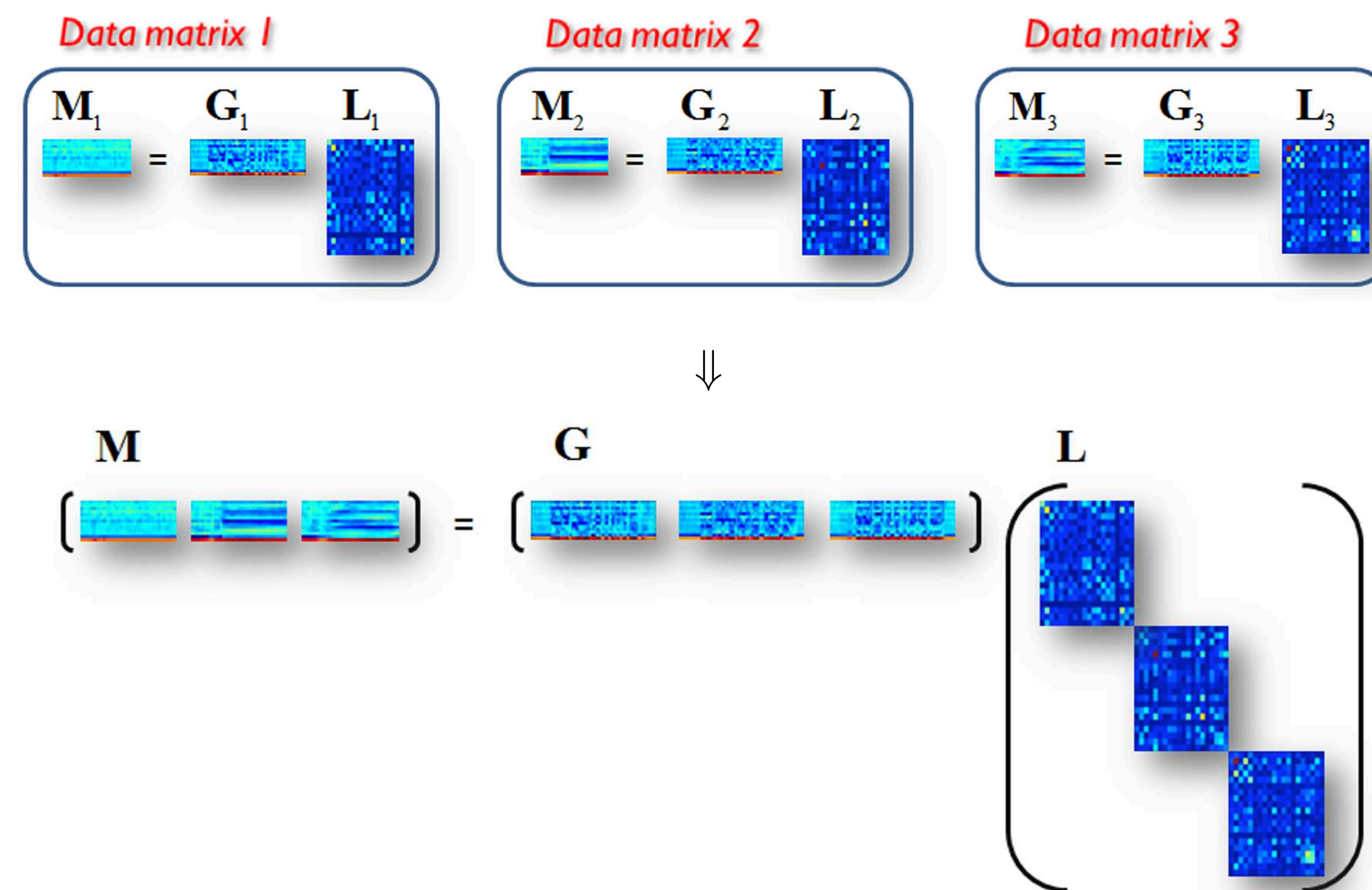
– SVT comprises of the two following iterative steps

$$\begin{cases} X_t = \mathcal{D}_\tau(\mathcal{A}_T^*(y_{t-1})) \\ y_t = y_{t-1} - \delta_t(\mathcal{A}_T(X_t) - \mathbf{b}) \end{cases}$$

- soft-thresholding operator $\mathcal{D}_\tau = U\Sigma_\tau V^T$, step size δ
 - $\Sigma_\tau = \text{diag}(\max\{\sigma_i - \tau, 0\})$
 - singular values $\{\sigma_i\}_{i=1}^{\min\{n_1, n_2\}}$

Joint low-rank representation and matrix completion

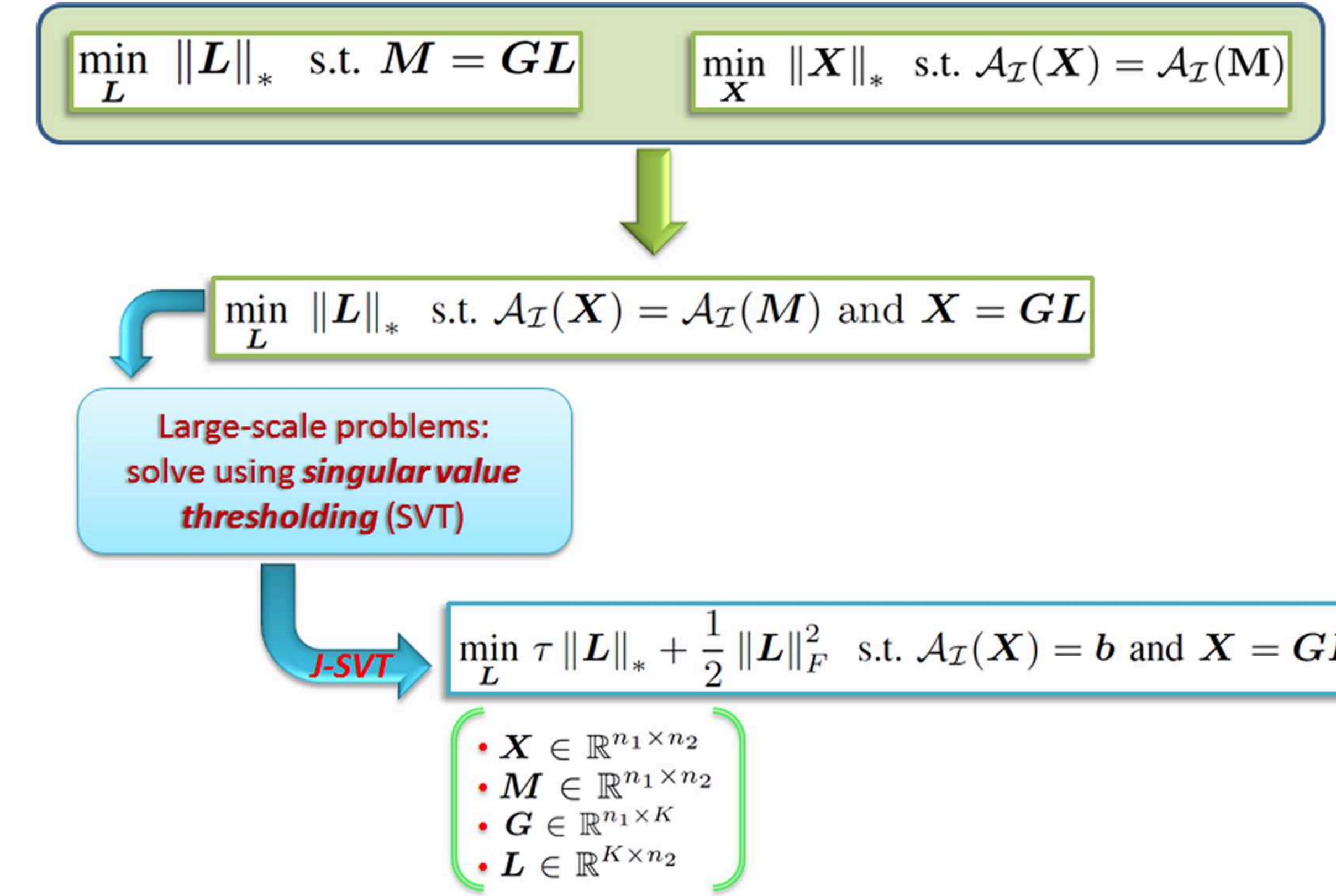
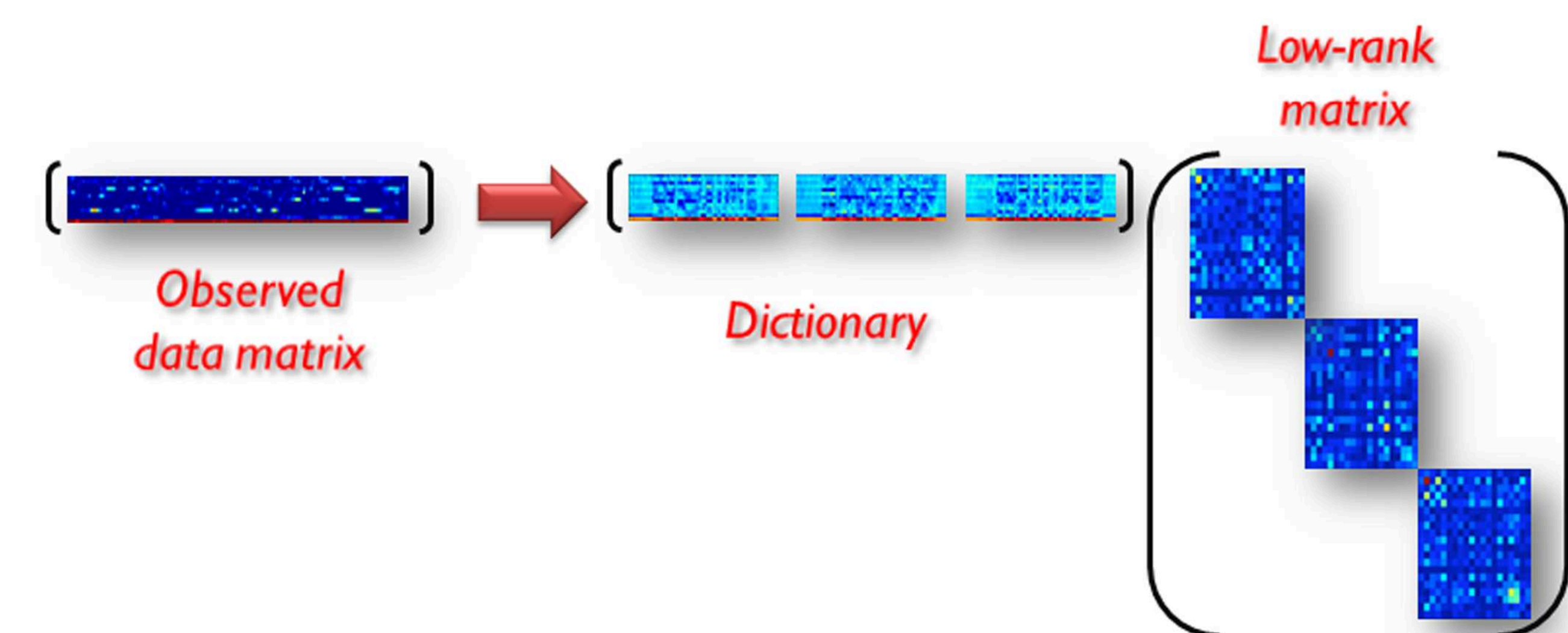
- Low-rank matrix completion $\implies$ unsupervised approach
 - exploit *a-priori* knowledge of data at hand $\implies$ promote a supervised version of matrix completion
- Typical low-rank matrix completion $\implies$...any information about the *procedure generating* the data matrix M ?
- NO!** ...introduce *low-rank representation*
 - exploit the supervised nature of a dictionary-based task



– block diagonal matrix L is the lowest rank representation of the original data matrix M with respect to the dictionary G

$$\min_L \|L\|_* \quad \text{s.t. } M = GL$$

- Apply low-rank representation on matrices with missing data
 - combine matrix completion and low-rank representation $\implies$ seeks a low-rank coefficient matrix L from a small number of measurements $\mathcal{A}_T(M)$



- Propose a joint low-rank representation and matrix completion version of SVT $\implies$ J-SVT

Algorithm 1: J-SVT algorithm
Input: $\mathcal{A}_T$, observed values $\mathbf{b}$, dictionary G , step size δ , tolerance ϵ , parameter $\tau > 0$, maximum iterations $t_{\max}$.
Output: estimated matrix $X = GL_T$.

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1 Initialization:  $y_t = \tau b / \|G^T \mathcal{A}_T^*(b)\|$ 
2 for  $t = 1$  to  $t_{\max}$  do
3    $[U_t, \Sigma_t, V_t, s_t] = \text{SVDshrink}(G^T \mathcal{A}_T^*(y_t), \tau, s_{t-1})$ 
4    $L_t = U_t \Sigma_t V_t^T$ 
5   if  $\|\mathcal{A}_T(GL_t) - \mathbf{b}\|_2 \leq \epsilon \|\mathbf{b}\|_2$  then
6     break
7    $y_{t+1} = y_t - \delta(\mathcal{A}_T(GL_t) - \mathbf{b})$ 
8 end
```

Algorithm 3: Accelerated J-SVT algorithm: identical to J-SVT except replace line 8 in J-SVT with the following and initialize $z_1 = y_1$.

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8  $z_{t+1} = y_t - \delta(\mathcal{A}_T(GL_t) - \mathbf{b})$ 
9  $y_{t+1} = z_{t+1} + \frac{t}{t+1}(z_{t+1} - z_t)$ 
```

Algorithm 2: SVDshrink algorithm
Input: internal integer parameter ℓ .
function $\text{SVDshrink}(Z, \tau, s_0)$
1 $s \leftarrow s_0 + 1$
2 repeat
3 $[U, \Sigma, V] = \text{partialSVD}(Z, s)$
4 $s \leftarrow s + \ell$
5 until $\Sigma_{s, s} \leq \tau$
6 return $[U, \mathcal{D}_\tau(\Sigma), V, s]$
end function

Experimental results

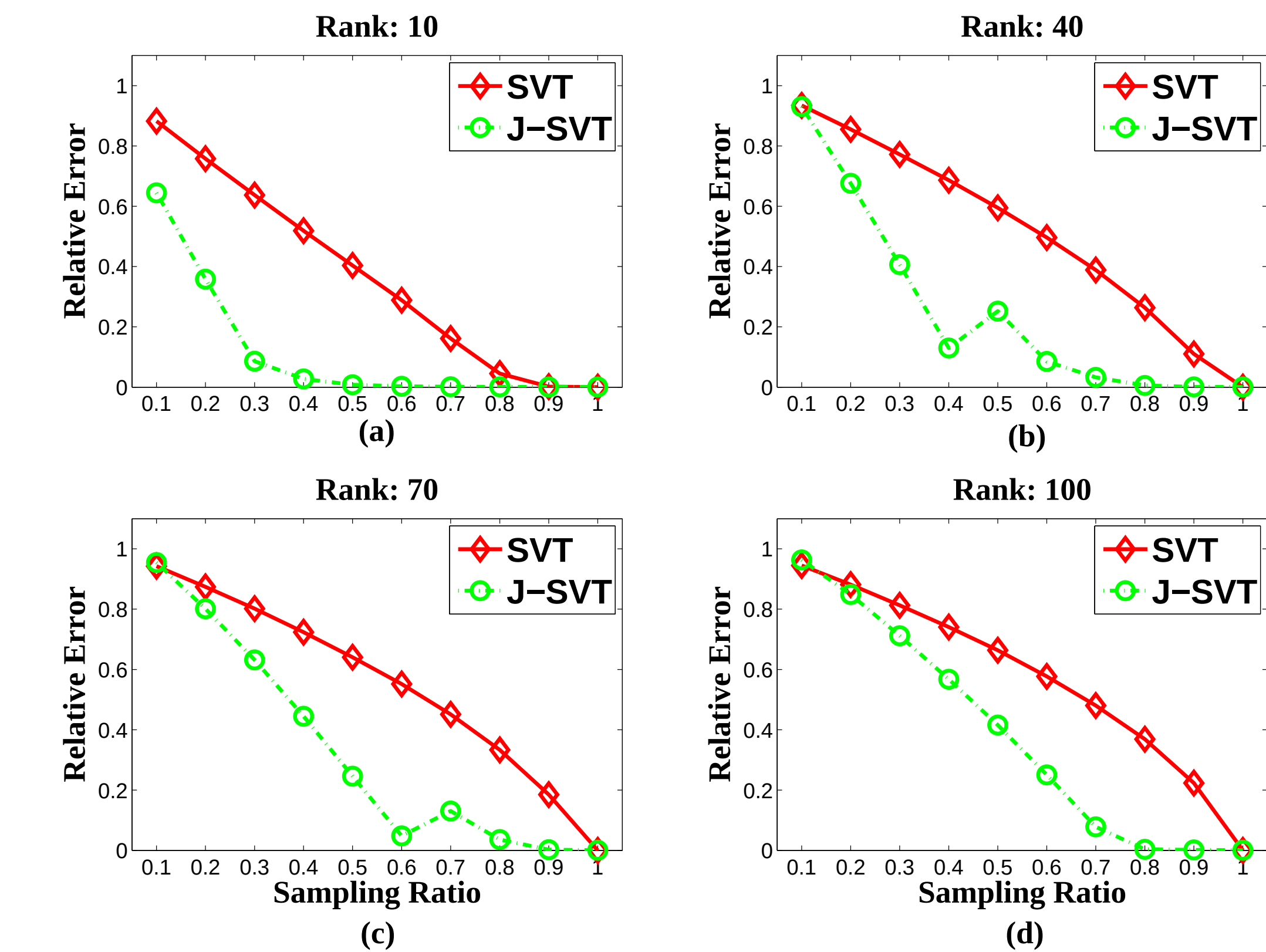
- Examine the reconstruction performance of J-SVT vs. SVT
 - simulations on synthetic data (normally distributed random samples)
 - dictionary G
 - low-rank representation matrix L
 - evaluation metric

$$\text{Relative error} = \frac{\|\hat{X} - M\|_F}{\|M\|_F}$$

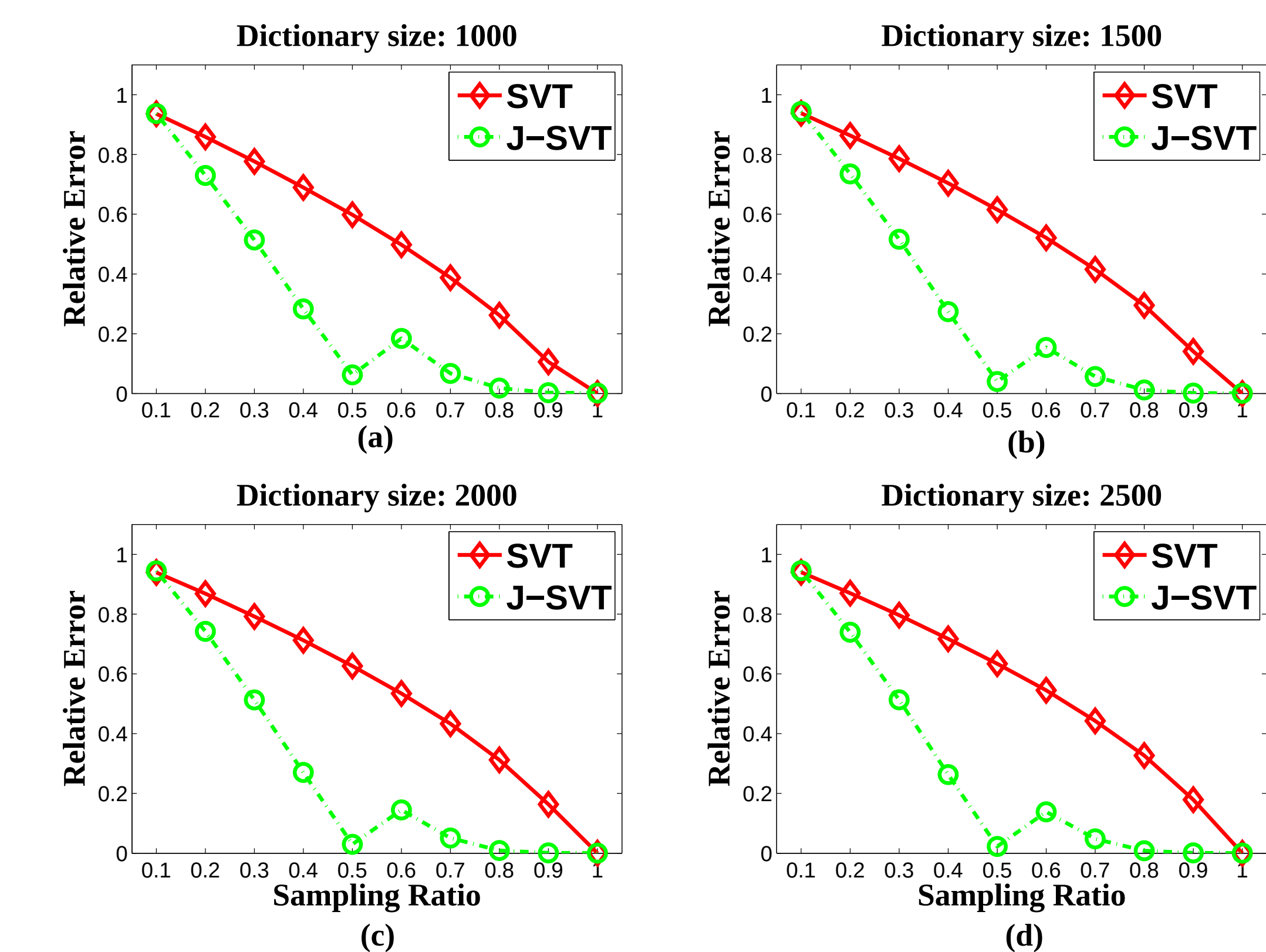
- Experimental parameters:**

- $n_1 \times n_2 = 300 \times 500$
- maximum number of iterations $t_{\max} = 100$
- tolerance $\epsilon = 10^{-5}$
- optimization parameter $\tau = 5\sqrt{n_1 n_2}$
- step size δ
 - SVT $\rightarrow 1.9$
 - J-SVT $\rightarrow \|G\|^{-2}$

- J-SVT reconstruction performance for a varying matrix rank
 - fixed dictionary size $\rightarrow 300 \times 1500$



- J-SVT reconstruction performance for a varying dictionary size
 - fixed rank(L) $\rightarrow 50$



- Robustness of J-SVT vs. SVT under noisy conditions

– fixed dictionary size $\rightarrow 300 \times 1500$
– fixed rank(L) $\rightarrow 50$

